

Is Stiffness Sufficient for Palpation Lesion Localization?

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Abstract—Robotic palpation often converts each indentation into an apparent-stiffness map, reducing a force–depth response to a single slope. This convenient scalarization hides loading history and makes the representation depend on the selected indentation depths. We ask whether complete displacement and F_z trajectories provide a more faithful input for single-probe deep-lesion localization. Using a controlled soft-body simulator, we generated soft phantoms with paired curve scans, stiffness maps, and projected inclusion masks, and compared matched U-Net models under fixed and variable indentation depths. Preserving the curve increased held-out Dice from 0.792 to 0.819 under fixed depths and from 0.573 to 0.712 when terminal depth varied across scan locations. Together, these results identify scalar stiffness as both an information bottleneck and a depth-pair-dependent measurement. These findings support process-level force sensing for simulated single-probe localization, while physical validation remains future work.

Index Terms—robotic palpation, lesion localization, soft-body simulation

I. INTRODUCTION

Palpation infers hidden tissue structure from mechanical response. Robotic palpation makes this interaction repeatable by controlling indentation and measuring force, but subsurface structure remains indirectly observed through the surface response.

Most pipelines do not retain this process. They compress each press into peak force, two-point equivalent stiffness, or a related scalar map. Such summaries are interpretable, but a single slope cannot represent nonlinear loading history, curve shape, or depth-dependent force changes. The value also changes when different indentation samples define the slope. This creates a representation problem: a scalar stiffness map may be useful as an engineering summary, yet insufficient as an input for localization. Figure 1 summarizes this representation choice.

We study this representation problem for single-probe normal-force palpation. A spherical probe presses a soft phantom on a regular grid and records displacement and F_z at each location. The task is to predict the 2D surface projection of embedded stiff inclusions. The fixed-depth benchmark asks whether scalar stiffness loses useful loading information under a shared indentation schedule. The random-depth benchmark asks whether the same scalar summary also degrades when the depth pair varies across scan locations.

The contributions are:

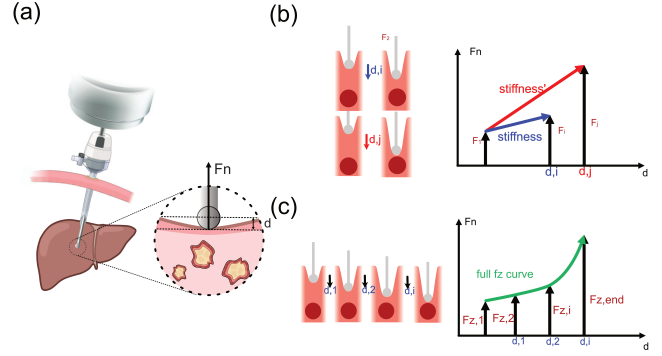


Fig. 1. Motivating representation choice. (a) A probe records normal force during indentation. (b) Two-point stiffness is a slope between selected force–depth samples. (c) The proposed curve input preserves sampled displacement and force across the press.

- 1) We formulate single-probe palpation as 2D mask prediction from scan-grid force–depth measurements.
- 2) We construct a VBD/FEM benchmark with paired displacement/ F_z curves, equivalent-stiffness maps, and projected inclusion masks.
- 3) We show, under matched U-Net comparisons, that displacement/ F_z curves outperform stiffness maps, with a larger advantage when indentation depth varies across scan locations.

II. RELATED WORK

Robotic palpation commonly converts contact response into an apparent-stiffness map. Early systems used these maps to localize stiff regions [1, 2], and later work improved sampling efficiency with Gaussian-process active search, level-set methods, and trajectory-optimized sensing [3, 4]. Recent exploratory systems continue to update stiffness maps for localization and contour following [5, 6]. These methods are practical, but the inference remains mediated by a scalar contact representation.

Several studies estimate boundaries or projected extents from stiffness-derived maps. Nichols and Okamura used supervised classifiers and MRF refinement [7], Yan and Pan combined Bayesian localization with sliding palpation [8], and recent RMIS pipelines segment tumors from reconstructed force or stiffness maps [9, 10, 11]. Encoder-decoder models such as U-Net are well suited to projected-mask prediction [12], but they are usually applied to processed maps rather than the full press sequence.

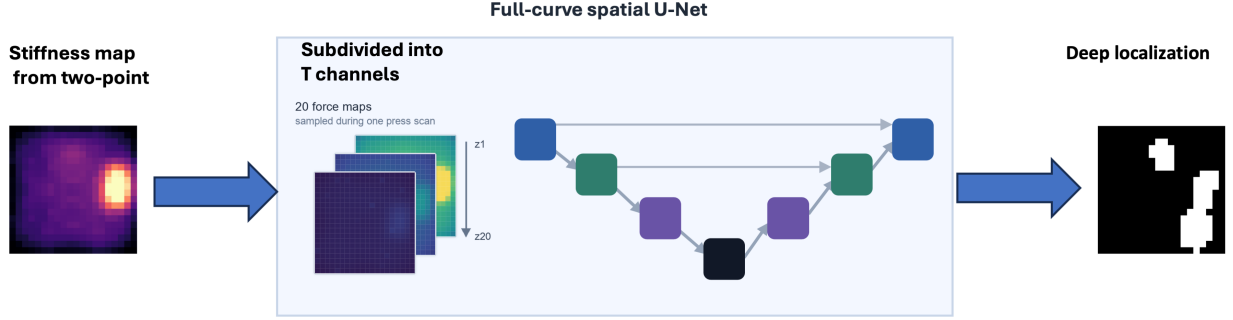


Fig. 2. Method overview. The conventional two-point analysis compresses the press response into a stiffness map, whereas the process-aware analysis represents the scan as a curve-channel input and supplies the stack to a spatial U-Net for deep-lesion localization. The schematic highlights the force-channel stack; the reported curve inputs use paired displacement and F_z samples.

Ultrasound elastography provides a related motivation. It treats deformation or strain as lesion evidence, and recent robotic work combines B-mode ultrasound with strain elastography for tumor localization [13]. Our image-free setting examines whether an analogous process cue exists in single-probe force–depth signals. Related tactile studies use force–displacement sequences for depth recognition [14, 15, 16], show that action changes tactile separability [17, 18], or learn representations from soft-body touch [19]. Richer tactile tomography reconstructs hidden fields from specialized sensing [20, 21]. We therefore focus on a narrower question: with scan geometry, model capacity, and mask objective fixed, does preserving the complete force–depth trajectory improve 2D localization over a two-point stiffness map?

III. PROBLEM AND DATASET

A palpation trial records a tensor $\mathbf{P} \in \mathbb{R}^{H \times W_g \times T \times 2}$ over a rectangular scan grid. At each grid point, the probe follows a normal indentation trajectory and records displacement and F_z . The target $\mathbf{M}$ is the projected binary inclusion mask over the scan area. In the reported high-resolution benchmark, inputs remain on the 20×20 scan grid and targets are sampled on a 128×128 grid. Table I summarizes the simulation recipe.

The dataset is generated by tetrahedral soft-body simulation in the Newton Physics Engine using its VBD solver and a Neo-Hookean-like material assignment [22, 23, 24]. We use the simulator as a controlled generator rather than a patient-specific model; the formulation follows standard deformable-solid FEM and nonlinear finite-element solution practice, with biomechanical FE precedent [25, 26, 27]. Each soft block contains one to four analytic stiff inclusions. Their geometry defines both the high-stiffness mesh region and the projected mask. The reported benchmark expands the data recipe to

TABLE I
SYNTHETIC DATASET RECIPE USED FOR THE SIMULATION BENCHMARK.

Item	Setting
Synthetic phantom	Soft block, $180 \times 180 \times 80$ mm
Synthetic scan	20×20 grid, 20 press steps
Synthetic inclusions	Random 1–4 stiff inclusions per phantom
Shape set	Sphere, ellipsoid, box, cylinder, capsule
Simulation model	VBD/FEM tetrahedral mesh, $48 \times 48 \times 16$ cells
Stiffness contrast	$20 \times$ base material in reported simulations
Material assignment	Tetrahedron centroids inside analytic inclusions
Measurement tensor	Displacement/ F_z array, $20 \times 20 \times 20 \times 2$
Segmentation label	Analytic projected mask, 128×128 grid
Benchmark split	1000 train / 400 validation / 400 test phantoms

1000 training phantoms, 400 validation phantoms, and 400 independent test phantoms drawn by random split from a 1800-phantom synthetic pool.

The fixed-depth benchmark uses a common indentation schedule at every scan location. It therefore isolates the information lost when a force–depth curve is compressed to one stiffness value. The random-depth benchmark chooses one reproducible random terminal depth per scan location. The stiffness input then uses a slope over a varying depth pair, whereas the curve input resamples the same truncated prefix to 20 displacement/ F_z samples.

IV. METHODS

Figure 2 summarizes the pipeline. The conventional two-point equivalent-stiffness representation reduces each press to one slope,

$$k_{i,j} = \frac{F_{z,i,j,T} - F_{z,i,j,1}}{d_{i,j,T} - d_{i,j,1}}, \quad (1)$$

where d is indentation displacement, F_z is normal force, and T is the final indentation sample. The random-depth benchmark evaluates the same two-point definition at the selected terminal sample for each scan location. The process representation instead stacks displacement and F_z samples as curve channels on the 20×20 scan grid. In the random-depth benchmark, the prefix from contact to the selected terminal depth is resampled to 20 steps, giving 40 input channels. Stiffness inputs remain one-channel 20×20 maps. The target mask is evaluated at 128×128 resolution, and any required logit resizing is performed inside the model path. The primary model is a compact 2D U-Net with approximately 1.10M trainable parameters. Using the same U-Net architecture across stiffness and curve inputs isolates the effect of input representation.

All neural models are trained with binary cross entropy plus soft Dice loss using AdamW, learning rate 10^{-3} , weight decay 10^{-4} , and batch size 16. All neural baselines are trained on 1000 phantoms for at most 50 epochs, with patience 15. Checkpoints are selected by the lowest validation loss under the same continuous training objective. The complete trained inference pipeline is then applied unchanged to the 400-phantom independent test split.

V. EXPERIMENTS

The main test results use the independent test phantoms from the VBD/FEM data generation pipeline. Model outputs are evaluated against the projected binary 128×128 mask. Each neural baseline is evaluated as a fixed trained pipeline applied unchanged to the independent test split. All confusion counts are computed over mask pixels after selecting the binarization threshold on the validation split. Here, TP denotes inclusion pixels predicted as inclusion, FP denotes background pixels predicted as inclusion, and FN denotes inclusion pixels missed by the prediction. Pixel-level F1 is algebraically equivalent to Dice for this binary mask task, so Dice is the primary metric:

$$\text{Dice} = \frac{2TP}{2TP + FP + FN}, \quad \text{IoU} = \frac{TP}{TP + FP + FN}. \quad (2)$$

VI. RESULTS

TABLE II
VALIDATION-SELECTED TEST DICE AND IOU ON 128×128 MASKS.

Setting	Model	Input	Dice	IoU
Fixed depth	Point MLP	Stiffness map	0.626	0.456
Fixed depth	Shallow CNN	Stiffness map	0.715	0.556
Fixed depth	U-Net	Stiffness map	0.792	0.656
Fixed depth	Point MLP	$F_z + d$ curve	0.617	0.446
Fixed depth	Shallow CNN	$F_z + d$ curve	0.783	0.643
Fixed depth	U-Net	$F_z + d$ curve	0.819	0.693
Random depth	U-Net	Stiffness map	0.573	0.402
Random depth	U-Net	$F_z + d$ curve	0.712	0.553

Table II reports held-out test Dice and IoU. In the fixed-depth benchmark, the matched U-Net improved from 0.792 Dice and 0.656 IoU with stiffness maps to 0.819 Dice and 0.693 IoU with displacement/ F_z curves. The shallow CNN showed a larger increase, from 0.715 to 0.783 Dice and from 0.556 to 0.643 IoU, whereas the pointwise MLP did not benefit from the curve input.

The random-depth benchmark showed a larger representation gap. With the same U-Net architecture, Dice and IoU were 0.573 and 0.402 for stiffness maps, and 0.712 and 0.553 for resampled curve inputs.

VII. DISCUSSION

The results support a representation claim rather than a general detector claim. When all presses share the same indentation schedule, the curve input preserves loading information that is absent from a terminal slope. When terminal depth varies, the stiffness map is also coupled to the depth pair used to compute it. The displacement/ F_z curve avoids reducing this variability to a single number, giving the model both force evolution and depth context.

The random-depth result also marks a boundary of the current method. In the reported model, displacement and F_z are concatenated as U-Net channels after resampling. This simple encoding improves robustness, but it leaves temporal alignment and physical depth normalization largely to the network. More principled depth encoding may be needed when moving from controlled simulation to physical palpation. The study also assumes flat surfaces, registered scan grids, controlled normal loading, and known projected masks. Real palpation will add contact-onset uncertainty, sensor noise, material mismatch, surface geometry, and finite sensing reach.

VIII. CONCLUSION

We formulated single-probe palpation as 2D segmentation over force–depth scan grids. Across fixed-depth and random-depth simulation benchmarks, matched U-Net comparisons favored displacement/ F_z curve inputs over stiffness maps. The random-depth setting further exposed the dependence of stiffness on depth-pair selection. These controlled comparisons support process-aware force sensing for deep-lesion localization, while physical validation under real contact variability remains future work.

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